

## B1 Fluids — Model Solutions, MT 2020

### Question 1 — Oscillating plate (Stokes' second problem)

**Problem.** Incompressible viscous fluid above an infinite plate at  $z = 0$  that oscillates as  $\mathbf{u}|_{z=0} = (U \cos \omega t, 0)$ ; velocity decays for  $z \rightarrow \infty$ . Gravity neglected.

#### (a) Governing equations

Incompressible Navier–Stokes (no gravity):

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}. \quad (1)$$

Mass conservation:

$$\nabla \cdot \mathbf{u} = 0.$$

Term-by-term:  $\partial_t \mathbf{u}$  local acceleration;  $(\mathbf{u} \cdot \nabla) \mathbf{u}$  advective (inertial) acceleration;  $-\rho^{-1} \nabla p$  pressure-gradient force;  $\nu \nabla^2 \mathbf{u}$  viscous diffusion of momentum;  $\nabla \cdot \mathbf{u} = 0$  enforces incompressibility.

#### (b) Velocity field

Translation invariance: plate is infinite in  $x$ , forcing is  $x$ -independent:

$$\mathbf{u} = [u_x(z, t), u_z(z, t)], \quad p = p(z, t).$$

Mass conservation:

$$\frac{\partial u_z}{\partial z} = 0.$$

Apply  $u_z \rightarrow 0$  as  $z \rightarrow \infty$ :

$$u_z \equiv 0.$$

Substitute into  $x$ -component of (1) (advection  $u_z \partial_z u_x = 0$ , no  $\partial_x p$ ):

$$\frac{\partial u_x}{\partial t} = \nu \frac{\partial^2 u_x}{\partial z^2}. \quad (2)$$

Try  $u_x = \text{Re}[u_0(z) e^{i\omega t}]$ :

$$i\omega u_0 = \nu u_0''.$$

Rearrange:

$$u_0'' = \frac{i\omega}{\nu} u_0.$$

Roots  $\pm k$  with  $k = \sqrt{i\omega/\nu} = (1+i)\sqrt{\omega/(2\nu)}$ . Decay as  $z \rightarrow \infty$  kills  $e^{+kz}$ :

$$u_0(z) = U e^{-kz}.$$

Apply  $u_x(0, t) = U \cos \omega t$ , fixing the amplitude. Define  $\delta = \sqrt{2\nu/\omega}$ :

$$\boxed{u_x(z, t) = U e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta}\right), \quad u_z = 0} \quad (3)$$

### (c) Stress tensor

Newtonian stress:

$$\sigma_{ij} = -p\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right).$$

Use  $u_z = 0$  and  $\partial_x u_x = 0$ :

$$\sigma_{xx} = -p, \quad \sigma_{zz} = -p.$$

Off-diagonal:

$$\sigma_{xz} = \sigma_{zx} = \mu \frac{\partial u_x}{\partial z}.$$

Differentiate (3):

$$\frac{\partial u_x}{\partial z} = -\frac{U}{\delta} e^{-z/\delta} [\cos(\omega t - \frac{z}{\delta}) - \sin(\omega t - \frac{z}{\delta})].$$

Apply  $\cos \alpha - \sin \alpha = \sqrt{2} \cos(\alpha + \pi/4)$ :

$$\frac{\partial u_x}{\partial z} = -\frac{\sqrt{2}U}{\delta} e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta} + \frac{\pi}{4}\right).$$

Therefore

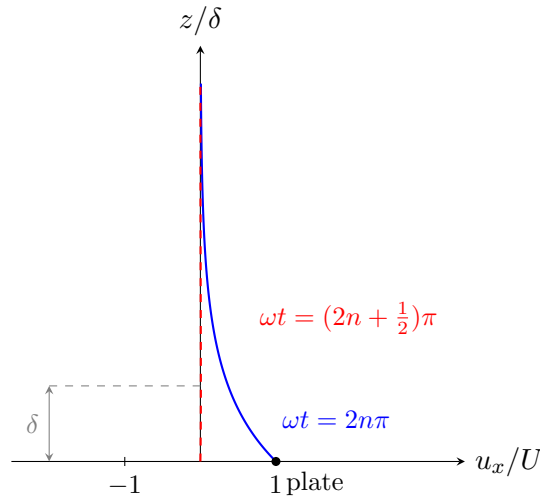
$$\sigma_{xz} = \sigma_{zx} = -\mu \frac{\sqrt{2}U}{\delta} e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta} + \frac{\pi}{4}\right), \quad \sigma_{xx} = \sigma_{zz} = -p$$

( $p$  undetermined to within an additive constant since  $\partial_z p = 0$  and gravity is neglected).

### (d) Profiles, force balance, boundary layer

Profiles:

$$u_x(z, t = 2n\pi/\omega) = U e^{-z/\delta} \cos(z/\delta), \quad u_x(z, t = (2n + \frac{1}{2})\pi/\omega) = U e^{-z/\delta} \sin(z/\delta).$$



Force balance: in (2) unsteady inertia  $\partial_t u_x$  balances viscous diffusion  $\nu \partial_z^2 u_x$ . Pressure is uniform; advection vanishes. Momentum injected at the plate diffuses upward while the imposed phase reverses every half period, generating the damped travelling wave (3).

Boundary-layer thickness from  $\delta = \sqrt{2\nu/\omega}$ :

$$\delta = \sqrt{2\nu/\omega}$$

Independent of  $U$ . Physical reading: in one forcing period  $T = 2\pi/\omega$  vorticity diffuses a distance  $\sqrt{\nu T} \sim \delta$  before the surface reverses, halting further penetration.

Tidal-flow application: by Galilean change of frame to one in which the fluid oscillates at  $-U \cos \omega t$  and the plate is fixed, the same equation governs flow over a stationary bed with oscillatory far-field velocity. So an oscillatory Stokes layer of thickness  $\delta = \sqrt{2\nu/\omega}$  does form near a tidal bed. Numerical estimate ( $\nu = 10^{-6} \text{ m}^2 \text{ s}^{-1}$ ,  $\omega = 1.4 \times 10^{-4} \text{ s}^{-1}$  semi-diurnal):  $\delta \sim 4 \text{ mm}$ . Caveats: real tidal flow is turbulent (large  $\text{Re}$ ), so the laminar Stokes layer is replaced by a thicker turbulent boundary layer; the laminar solution describes only the viscous sub-layer.

## Question 2 — Active cylinder, potential flow

**Problem.** 2D inviscid, incompressible, irrotational flow past cylinder  $r = a$  with prescribed normal velocity  $u_r(a, \theta) = A \sin \theta$ , far-field  $\mathbf{u} \rightarrow -U \hat{\mathbf{x}}$ ,  $p \rightarrow p_0$ . Cylinder axis vertical.

### (a) Velocity potential and Bernoulli

Irrotational:  $\nabla \times \mathbf{u} = 0$ , so  $\mathbf{u} = \nabla \phi$ . Incompressible:  $\nabla \cdot \mathbf{u} = 0$ , hence

$$\nabla^2 \phi = 0.$$

Euler with gravity:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p - g \hat{\mathbf{z}}.$$

Vector identity:

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \left( \frac{1}{2} |\mathbf{u}|^2 \right) - \mathbf{u} \times (\nabla \times \mathbf{u}).$$

Apply  $\nabla \times \mathbf{u} = 0$  and  $\mathbf{u} = \nabla \phi$ :

$$\nabla \left( \frac{\partial \phi}{\partial t} + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho} + gz \right) = 0.$$

Therefore  $\nabla H = 0$  with

$$\boxed{H = \frac{\partial \phi}{\partial t} + \frac{1}{2} |\mathbf{u}|^2 + \frac{p}{\rho} + gz} \quad (4)$$

Steady interpretation:  $\partial_t \phi = 0$ , so  $H$  is the conserved total mechanical energy per unit mass — kinetic + pressure-work term + gravitational potential.

### (b) Solution and surface pressure

BCs:

$$\left. \frac{\partial \phi}{\partial r} \right|_{r=a} = A \sin \theta, \quad \nabla \phi \rightarrow -U \hat{\mathbf{x}} \text{ as } r \rightarrow \infty.$$

Resolve  $-U \hat{\mathbf{x}} = -U \cos \theta \hat{\mathbf{r}} + U \sin \theta \hat{\boldsymbol{\theta}}$ :

$$\phi \rightarrow -Ur \cos \theta \quad (r \rightarrow \infty).$$

From the general expansion keep  $n = 1$  modes only:

$$\phi = \left( A_1 r + \frac{B_1}{r} \right) \cos \theta + \left( C_1 r + \frac{D_1}{r} \right) \sin \theta.$$

Match far field:

$$A_1 = -U, \quad C_1 = 0.$$

Apply  $\partial_r \phi|_a = A \sin \theta$ :

$$\left(-U - \frac{B_1}{a^2}\right) \cos \theta + \left(-\frac{D_1}{a^2}\right) \sin \theta = A \sin \theta.$$

Match coefficients:

$$B_1 = -Ua^2, \quad D_1 = -Aa^2.$$

Hence

$$\boxed{\phi(r, \theta) = -U \left(r + \frac{a^2}{r}\right) \cos \theta - \frac{Aa^2}{r} \sin \theta} \quad (5)$$

Velocity:

$$u_r = \frac{\partial \phi}{\partial r} = -U \left(1 - \frac{a^2}{r^2}\right) \cos \theta + \frac{Aa^2}{r^2} \sin \theta,$$

$$u_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = U \left(1 + \frac{a^2}{r^2}\right) \sin \theta - \frac{Aa^2}{r^2} \cos \theta.$$

At  $r = a$ :

$$u_r(a, \theta) = A \sin \theta, \quad u_\theta(a, \theta) = 2U \sin \theta - A \cos \theta.$$

Surface speed squared:

$$|\mathbf{u}(a, \theta)|^2 = A^2 \sin^2 \theta + (2U \sin \theta - A \cos \theta)^2.$$

Expand:

$$|\mathbf{u}|^2 = A^2 \sin^2 \theta + 4U^2 \sin^2 \theta - 4UA \sin \theta \cos \theta + A^2 \cos^2 \theta.$$

Combine  $A^2 \sin^2 \theta + A^2 \cos^2 \theta = A^2$  and  $2 \sin \theta \cos \theta = \sin 2\theta$ :

$$|\mathbf{u}|^2 = A^2 + 4U^2 \sin^2 \theta - 2UA \sin 2\theta.$$

Steady Bernoulli on the level  $z = \text{const}$  around the cylinder (axis vertical, gravity drops):

$$p + \frac{1}{2} \rho |\mathbf{u}|^2 = p_0 + \frac{1}{2} \rho U^2.$$

Solve for  $p(a, \theta)$ :

$$p(a, \theta) = p_0 + \frac{1}{2} \rho U^2 - \frac{1}{2} \rho [A^2 + 4U^2 \sin^2 \theta - 2UA \sin 2\theta].$$

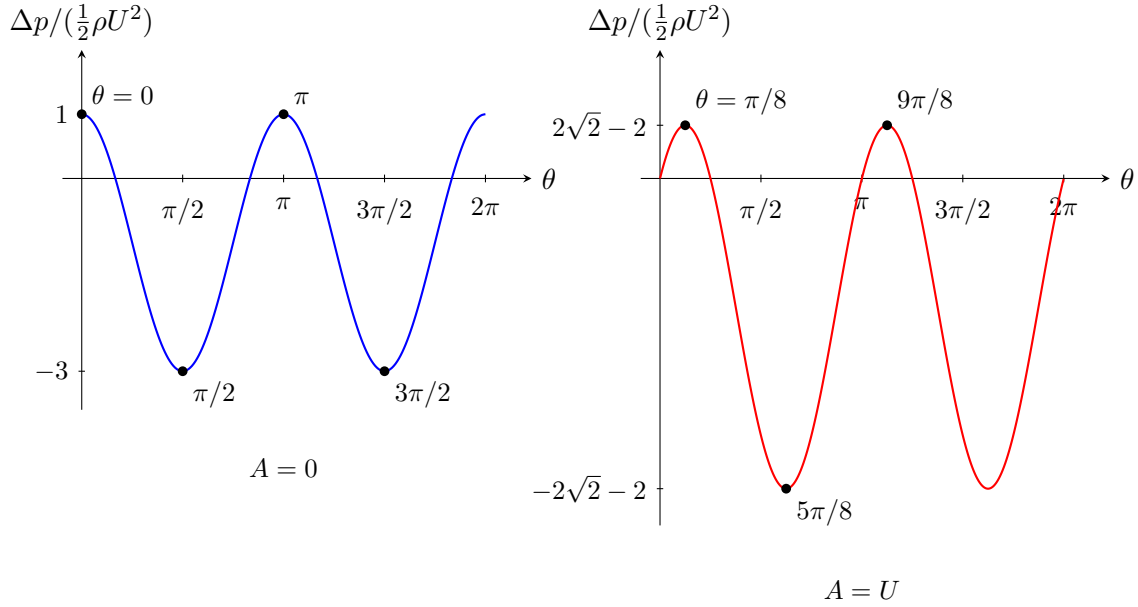
Use  $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ :

$$p(a, \theta) = p_0 + \frac{1}{2} \rho U^2 - \frac{1}{2} \rho A^2 - \rho U^2 (1 - \cos 2\theta) + \rho U A \sin 2\theta.$$

Combine  $\frac{1}{2} \rho U^2 - \rho U^2 = -\frac{1}{2} \rho U^2$ :

$$\boxed{p(a, \theta) = p_0 + \frac{1}{2} \rho U^2 \left[ 2 \cos 2\theta - 1 - \frac{A^2}{U^2} + 2 \frac{A}{U} \sin 2\theta \right]} \quad (6)$$

Sketch of  $\Delta p \equiv p(a, \theta) - p_0$  in units of  $\frac{1}{2} \rho U^2$ :



For  $A = 0$ , fore-aft symmetric about  $\theta = \pi/2, 3\pi/2$ ; for  $A = U$ , the curve shifts and tilts: maximum suction at  $\theta = 5\pi/8$ , peak pressure at  $\theta = \pi/8$  — a transverse asymmetry that lifts the cylinder.

### (c) Boundary-layer separation

Inert case  $A = 0$ . Inviscid surface speed  $u_\theta = 2U \sin \theta$ , so a no-slip boundary layer grows everywhere except the stagnation points  $\theta = 0, \pi$ . Surface pressure gradient (using (6) with  $A = 0$ ):

$$\frac{1}{a} \frac{\partial p}{\partial \theta} = -\frac{2\rho U^2}{a} \sin 2\theta.$$

Adverse on the rear half,  $\pi/2 < \theta < \pi$  (and mirror  $\pi < \theta < 3\pi/2$ ). Separation occurs shortly after  $\theta = \pi/2$  on top and shortly after  $\theta = 3\pi/2$  on the bottom (rear shoulders).

Active case  $A = U$ , upper surface  $\theta > 0$ . Pressure gradient:

$$\frac{1}{a} \frac{\partial p}{\partial \theta} = \rho U^2 [-2 \sin 2\theta + 2 \cos 2\theta] \frac{1}{a}.$$

Adverse where  $\cos 2\theta < \sin 2\theta$ , i.e.  $\theta > \pi/8$ . Compared with  $A = 0$  the adverse-pressure region begins later in the inviscid pressure pattern *but* the new minimum is at  $5\pi/8$ , past the top. More importantly the imposed  $u_r = A \sin \theta > 0$  blows fluid *outward* from the wall on  $\theta > 0$ , energising the boundary layer (analogous to suction-blowing control). The active blowing therefore *inhibits* separation on the upper surface relative to  $A = 0$ .

## Question 3 — Vorticity dynamics and shear-layer mode

**Problem.** Vorticity equation, piecewise-shear base flow, linear interfacial mode.

### (a) Vorticity equation

Take curl of incompressible Navier–Stokes:

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \nabla \times [(\mathbf{u} \cdot \nabla) \mathbf{u}] = \nu \nabla^2 \boldsymbol{\omega}.$$

Identity:

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla(\tfrac{1}{2}|\mathbf{u}|^2) - \mathbf{u} \times \boldsymbol{\omega}.$$

Take curl, use  $\nabla \times \nabla(\cdot) = 0$  and  $\nabla \cdot \mathbf{u} = \nabla \cdot \boldsymbol{\omega} = 0$ :

$$\nabla \times (\mathbf{u} \times \boldsymbol{\omega}) = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} - (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}.$$

Combine:

$$\boxed{\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \nabla^2 \boldsymbol{\omega}} \quad (7)$$

Term-by-term:  $D\boldsymbol{\omega}/Dt$  material rate of vorticity change following a fluid parcel;  $(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$  vortex stretching/tilting by the velocity gradient;  $\nu \nabla^2 \boldsymbol{\omega}$  viscous diffusion of vorticity.

2D inviscid:  $\boldsymbol{\omega} = \omega_z(x, y, t) \hat{\mathbf{z}}$ , no  $z$ -gradients in  $\mathbf{u}$ :

$$(\boldsymbol{\omega} \cdot \nabla) \mathbf{u} = \omega_z \partial_z \mathbf{u} = 0.$$

Drop viscosity:

$$\frac{D\omega_z}{Dt} = 0.$$

Streamfunction:  $u_x = \partial_y \psi$ ,  $u_y = -\partial_x \psi$ , so  $\omega_z = \partial_x u_y - \partial_y u_x = -\nabla^2 \psi$ :

$$\frac{D}{Dt} \nabla^2 \psi = 0. \quad (8)$$

## (b) Base flow

From  $\bar{U} = \partial_y \bar{\psi}$ :

$$\bar{\psi}(y) = \int_0^y \bar{U}(y') dy'.$$

Piecewise integration:

$$\bar{\psi}(y) = \begin{cases} -\frac{1}{2} \alpha y^2, & y \geq 0, \\ 0, & y < 0. \end{cases}$$

Vorticity  $\bar{\omega}_z = -\partial_y^2 \bar{\psi}$ :

$$\boxed{\bar{\omega}_z(y) = \begin{cases} \alpha, & y > 0, \\ 0, & y < 0. \end{cases}}$$

Each layer has uniform  $\bar{\omega}_z$ , and the base flow is steady and parallel:  $D(\nabla^2 \bar{\psi})/Dt = \bar{U} \partial_x (\nabla^2 \bar{\psi}) = 0$ , satisfying (8).

## (c) Linearised mode

Total  $\psi = \bar{\psi}(y) + \text{Re}[\hat{\psi}(y)e^{ikx-i\omega t}]$ . Linearise (8):

$$\left( \frac{\partial}{\partial t} + \bar{U} \frac{\partial}{\partial x} \right) \nabla^2 \psi' - (\nabla^2 \bar{\psi})_{,y} \frac{\partial \psi'}{\partial x} = 0.$$

Within each layer  $\bar{U}'' = 0$ , so:

$$(-i\omega + ik\bar{U})(-k^2 \hat{\psi} + \hat{\psi}'') = 0.$$

Cancel  $\omega - k\bar{U} \neq 0$ :

$$\boxed{\hat{\psi}'' - k^2 \hat{\psi} = 0} \quad (9)$$

Solutions decaying as  $|y| \rightarrow \infty$  (take  $k > 0$ ):

$$\hat{\psi}(y) = \begin{cases} C_+ e^{-ky}, & y > 0, \\ C_- e^{+ky}, & y < 0. \end{cases}$$

Pressure from horizontal momentum (linearised,  $u'_x = \partial_y \psi'$ ,  $u'_y = -\partial_x \psi'$ ):

$$\frac{\partial u'_x}{\partial t} + \bar{U} \frac{\partial u'_x}{\partial x} + u'_y \frac{\partial \bar{U}}{\partial y} = -\frac{1}{\rho} \frac{\partial p'}{\partial x}.$$

Substitute mode form  $u'_x = \hat{\psi}' e^{i(kx - \omega t)}$ ,  $u'_y = -ik \hat{\psi} e^{i(kx - \omega t)}$ :

$$(-i\omega + ik\bar{U})\hat{\psi}' + (-ik\hat{\psi})\bar{U}' = -\frac{ik}{\rho}\hat{p}.$$

Solve for  $\hat{p}$ :

$$\boxed{\hat{p} = \rho \left[ \frac{\omega - k\bar{U}}{k} \frac{\partial \hat{\psi}}{\partial y} + \hat{\psi} \frac{\partial \bar{U}}{\partial y} \right]} \quad (10)$$

Interface matching at  $y = \eta$ . Continuity of normal velocity / displacement gives  $[\hat{\psi}]_+^+ = 0$  at  $y = 0$ . Continuity of pressure (no surface tension) gives, using (10) at  $\bar{U}(0) = 0$ :

$$\left[ \frac{\omega}{k} \frac{\partial \hat{\psi}}{\partial y} + \hat{\psi} \frac{\partial \bar{U}}{\partial y} \right]_+^+ = 0 \quad \text{at } y = 0.$$

Apply continuity of  $\hat{\psi}$ :

$$C_+ = C_- \equiv C.$$

Evaluate the derivative jump. At  $y = 0^+$ :  $\hat{\psi}' = -kC$ ,  $\bar{U}' = -\alpha$ . At  $y = 0^-$ :  $\hat{\psi}' = +kC$ ,  $\bar{U}' = 0$ . Insert into the second condition:

$$\left[ \frac{\omega}{k}(-kC) + C(-\alpha) \right] - \left[ \frac{\omega}{k}(+kC) + 0 \right] = 0.$$

Collect terms:

$$-\omega C - \alpha C - \omega C = 0.$$

Simplify:

$$-2\omega C = \alpha C.$$

Therefore (with  $C \neq 0$ ):

$$\boxed{\omega = -\frac{\alpha}{2}} \quad (11)$$

Streamfunction (to within multiplicative constant):

$$\boxed{\hat{\psi}(y) = C e^{-k|y|}}$$

Evolution.  $\omega$  is real  $\Rightarrow$  disturbance is neutrally stable: amplitude constant in time. Phase factor  $e^{ikx - i\omega t} = e^{ik(x + \alpha t/(2k))}$ , giving phase speed  $c = \omega/k = -\alpha/(2k)$  in  $\hat{\mathbf{x}}$  (wave travels in  $-x$  for  $\alpha > 0$ ). Spatial structure:  $\hat{\psi} \propto e^{-k|y|}$ , decays exponentially away from the interface — a localised vorticity-edge wave on the kink in  $\bar{U}(y)$ .

## Question 4 — Lubrication: rotating cylinder near a plane

**Problem.** Long rigid cylinder, axis  $\parallel \hat{\mathbf{y}}$ , radius  $a$ , rotating  $\Omega \hat{\mathbf{y}}$  above plane  $z = 0$ , immersed in incompressible viscous fluid.

### (a) Reynolds number

$$\text{Re} = \frac{UL}{\nu} = \frac{\rho UL}{\mu}$$

ratio of inertial to viscous forces. High  $\text{Re} \gg 1$ : inertia dominates, viscous effects confined to thin boundary layers and wakes; turbulence possible. Low  $\text{Re} \ll 1$  (Stokes flow): viscous balance, time-reversible, flow  $\nabla^2 \mathbf{u} = \nabla p / \mu$  linear in boundary forcing, no separation, no inertia.

### (b)(i) Geometry, BCs, surface velocity

Cylinder axis at  $z = a + h_0$ . Surface:  $z_{\text{cyl}} = a + h_0 - \sqrt{a^2 - x^2}$ . Gap height:

$$h(x) = z_{\text{cyl}}(x) = h_0 + a - \sqrt{a^2 - x^2}.$$

Binomial expand for  $x \ll a$ :

$$\sqrt{a^2 - x^2} = a \sqrt{1 - x^2/a^2} \approx a - \frac{x^2}{2a}.$$

Substitute:

$$h(x) \approx h_0 + \frac{x^2}{2a} \quad (12)$$

Boundary conditions. No-slip + impermeability on rigid surfaces:

$$\mathbf{u}(z = 0) = \mathbf{0}, \quad \mathbf{u}|_{\text{cyl surface}} = \Omega \hat{\mathbf{y}} \times \mathbf{r}_{\text{from axis}}.$$

Position of contact-region surface point relative to axis:

$$\mathbf{r}_{\text{from axis}} = x \hat{\mathbf{x}} - (\sqrt{a^2 - x^2}) \hat{\mathbf{z}}.$$

Cross product:

$$\Omega \hat{\mathbf{y}} \times \mathbf{r}_{\text{from axis}} = \Omega \left[ \sqrt{a^2 - x^2} \hat{\mathbf{x}} + x \hat{\mathbf{z}} \right].$$

Apply  $x \ll a$ :

$$\mathbf{u}(x, z = h(x)) \approx U \hat{\mathbf{x}}, \quad U = \Omega a$$

( $\hat{\mathbf{z}}$  component is  $O(\Omega x) \ll U$ ).

### (b)(ii) Lubrication equation

Translation invariance: cylinder is long,  $\partial_y \equiv 0$ ,  $u_y \approx 0$ . Aspect ratio  $\epsilon \equiv h_0/a \ll 1$ . From  $\nabla \cdot \mathbf{u} = 0$ :

$$u_z \sim \epsilon u_x.$$

Compare derivatives:

$$\frac{\partial^2 u_x}{\partial x^2} \sim u_x/a^2, \quad \frac{\partial^2 u_x}{\partial z^2} \sim u_x/h_0^2, \quad \text{ratio} = \epsilon^2 \ll 1.$$



Drop  $\partial_x^2 u_x$ . Reduced Reynolds number:

$$\frac{\rho u_x \partial_x u_x}{\mu \partial_z^2 u_x} \sim \frac{\rho U^2 / a}{\mu U / h_0^2} = \frac{\rho U h_0^2}{\mu a} = \text{Re } \epsilon^2 \ll 1.$$

Drop inertia. Required:  $h \ll a$  and  $\rho U h^2 / (\mu a) \ll 1$ . The  $x$ -momentum equation reduces to

$$\boxed{\mu \frac{\partial^2 u_x}{\partial z^2} \approx -\frac{\partial p}{\partial x}} \quad (13)$$

Neglect  $\partial_x p$  as instructed:

$$\frac{\partial^2 u_x}{\partial z^2} = 0.$$

Integrate twice in  $z$ :

$$u_x(x, z) = B(x)z + C(x).$$

Apply  $u_x(z = 0) = 0$ :

$$C(x) = 0.$$

Apply  $u_x(z = h) = \Omega a$ :

$$B(x) = \frac{\Omega a}{h(x)}.$$

Therefore

$$\boxed{u_x(x, z) \approx \frac{\Omega a z}{h(x)}} \quad (14)$$

(linear Couette across the gap, shear rate  $\Omega a / h(x)$ ).

### (c) Force and torque

Viscous shear stress on the cylinder (in  $\hat{\mathbf{x}}$ ):

$$\tau_{xz}|_{z=h} = \mu \left. \frac{\partial u_x}{\partial z} \right|_{z=h} = \frac{\mu \Omega a}{h(x)}.$$

Net horizontal force on the cylinder (length  $L$ , integrated over the thin-gap strip  $|x| < b$ , sign such that the fluid drags the cylinder in  $-\hat{\mathbf{x}}$ ):

$$F \approx - \int_{-b}^b \frac{\mu \Omega a}{h(x)} L \, dx = \int_{-b}^b \frac{A}{h} \, dx, \quad A = -\mu \Omega a L.$$

Torque about the axis (lever arm  $\approx a$ ):

$$\Gamma \approx - \int_{-b}^b \frac{\mu \Omega a^2 L}{h(x)} \, dx = \int_{-b}^b \frac{B'}{h} \, dx, \quad B' = -\mu \Omega a^2 L.$$

Evaluate the integral with (12) extended to  $b \rightarrow \infty$ :

$$\int_{-\infty}^{\infty} \frac{dx}{h_0 + x^2 / (2a)}.$$

Substitute  $x = \sqrt{2ah_0} \tan \theta$ ,  $dx = \sqrt{2ah_0} \sec^2 \theta \, d\theta$ ,  $h = h_0 \sec^2 \theta$ :

$$\int_{-\infty}^{\infty} \frac{dx}{h(x)} = \frac{\sqrt{2ah_0}}{h_0} \int_{-\pi/2}^{\pi/2} d\theta = \sqrt{\frac{2a}{h_0}} \pi.$$

Therefore

$$\boxed{F \approx -\pi\mu\Omega a L \sqrt{\frac{2a}{h_0}}, \quad \Gamma \approx -\pi\mu\Omega a^2 L \sqrt{\frac{2a}{h_0}}} \quad (15)$$

Both diverge as  $h_0 \rightarrow 0$  — characteristic lubrication amplification.

#### (d) Free roll-and-slide

If the cylinder also translates with  $V(t)\hat{\mathbf{x}}$ , the surface point at the bottom moves at

$$\mathbf{v}_{\text{surf}} = V\hat{\mathbf{x}} + \Omega\hat{\mathbf{y}} \times (-a\hat{\mathbf{z}}) = (V - \Omega a)\hat{\mathbf{x}}.$$

The plate is fixed: top BC becomes  $u_x(z = h) = V - \Omega a$ . The lubrication analysis is linear, so replace  $\Omega a \rightarrow \Omega a - V$  in (c):

$$F \approx -\pi\mu L \sqrt{\frac{2a}{h_0}} (\Omega a - V),$$

$$\Gamma \approx -\pi\mu L a \sqrt{\frac{2a}{h_0}} (\Omega a - V).$$

Define

$$\lambda \equiv \pi\mu L \sqrt{2a/h_0}.$$

Equations of motion (slowly varying, no fluid inertia):

$$M \frac{dV}{dt} = F = -\lambda(\Omega a - V), \quad I \frac{d\Omega}{dt} = \Gamma = -\lambda a(\Omega a - V).$$

Track the slip  $s \equiv \Omega a - V$ :

$$\frac{ds}{dt} = a \frac{d\Omega}{dt} - \frac{dV}{dt}.$$

Substitute equations of motion:

$$\frac{ds}{dt} = -\frac{\lambda a^2}{I} s - \frac{\lambda}{M} s.$$

Combine:

$$\frac{ds}{dt} = -\frac{1}{\tau} s, \quad \frac{1}{\tau} = \lambda \left( \frac{a^2}{I} + \frac{1}{M} \right).$$

Integrate:

$$s(t) = s(0) e^{-t/\tau} \xrightarrow[t \rightarrow \infty]{} 0.$$

Therefore

$$\boxed{V(t \rightarrow \infty) = \Omega(t \rightarrow \infty) a}$$

Pure rolling. Combination  $MV + I\Omega/a$  is a conserved Casimir of the dynamics (no torque about the contact line from the lubrication shear when  $s = 0$ ).